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Aquatic Animal Health Package of Practices: Fish epidemiology and health economics



Fish epidemiology and health economics



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The absence of accurate baseline data on fish health and production, farm management practices, disease occurrence and mortalities, and the economic value of disease-related losses hampers disease control efforts by farmers and national governments. To address this, WorldFish and the Norwegian Veterinary Institute developed an open access survey tool to collect epidemiological data from different geographies, economies, and social and agro-climatic conditions. This online mobile phone data collection tool for farmer surveys generates the data needed for an integrated assessment of tilapia, carp and catfish epidemiology and health economics in Asia and Africa. The online tool is intended to replace the traditional paper-form survey and as a means of compiling baseline data more efficiently. Understanding fish epidemiology and health

economics is necessary to sustain small-scale and industrial aquaculture in the future to ensure the supply of safe, nutritious and affordable aquatic foods to nourish populations.

This package of practices includes the following key resources with links:

- The Fish Epidemiology and Health Economics survey tool: <https://bit.ly/3Azq1eO>
- A course on aquaculture survey with ODK developed on the Learn.ink online platform for digitized learning and training: <https://bit.ly/3AYXUGQ>
- A step-by-step guide to conducting digital surveys for aquaculture performance assessment: <https://hdl.handle.net/20.500.12348/4451>
- Content creator guidelines: How to manage digital surveys for aquaculture using KoBoToolbox and ODK Collect mobile app: <https://hdl.handle.net/20.500.12348/4892>

- End user guidelines: How to access digital surveys for aquaculture using ODK Collect and KoBoToolbox: <https://hdl.handle.net/20.500.12348/4893>
- End user guidelines: How to sign up and access the course on aquaculture survey with ODK on Learn.ink: <https://hdl.handle.net/20.500.12348/4894>

Our digital FEHE survey tool can collect a wide range of data, such as farm characteristics (e.g. location, size, production type and water source), pond preparation, stocking of species (e.g. numbers, weight and length at stocking and harvesting), biosecurity measures, source and use of inputs (e.g. seeds, feeds, chemicals, antibiotics and fertilizers), losses caused by poor management practices, losses caused by diseases, and the economic impacts of losses.

The tool comes with pre-online course modules developed on the Learn.ink platform. This allows for long-distance training and learning for our national partners, including university researchers and enumerators, aquaculture extension workers and government and non-governmental field officers. A step-by-step guide to prepare and conduct epidemiological surveys with aquaculture producers (link) is also part of the package.

Stakeholders, such as the competent authorities, can easily contextualize our digital epidemiology survey tool for wider adoption. They can choose the type of data they wish to collect and use their survey findings to identify risks, understand the performance of aquatic food systems and develop targeted interventions and national biosecurity strategies to support sustainable aquaculture development in their countries.

For scaling purposes, WorldFish will continue to promote uptake and use of this package of practices by national partners in our focal and scaling countries, in addition to bilateral projects presently being implemented by WorldFish and partners (e.g. the United States Agency for International Development Feed the Future Innovation Lab for Fish (USAID FIL) project in Nigeria, Agence Française de Développement (Afd) projects in Malawi and Zambia, Centre for Environment Fisheries and Aquaculture Science (Cefas) antimicrobial resistance and CGIAR COVID-19 country projects in Bangladesh and the Norwegian Agency for Development Cooperation (Norad) project in Egypt, Ghana and Kenya). This approach will ensure wider use of the fish epidemiology and health economics tool for collection of large data sets from different farming systems and geographies. Analysis of this data would lead to risk factors identification and development of risk mitigation interventions (better management practices), enabling sustainable aquatic food systems.

This package of practices is the first version and is a work in progress. WorldFish will continue to refine and revise the working versions based on feedback from value chain actors and lessons learned from the field in order to make it relevant to the national competent authorities in the countries where we work.

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